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When AI explains in natural language: Unveiling the impact of generative AI explanations on educators' grading and feedback practices

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Abstract

Grading and feedback provision for students' open-ended responses are time-consuming and cognitively demanding. Despite the advocacy to leverage Artificial Intelligence (AI) to automate assessment, concerns persist regarding inaccurate AI assessment on student learning and the potential detrimental effects on educators' assessment practices due to reliance. One alternative is to leverage AI-powered assessment insights as auxiliary information to support educators' assessment instead of solely entrusting AI for assessment. However, insights published in existing literature typically required excessive cognitive efforts from human graders to interpret and make use of. Generative AI (GenAI) technologies' capability To produce natural-language assessment explanations could potentially tackle this challenge. To empirically examine the efficacy of such natural-language insights while accounting for concerns in over-reliance on AI, we invited 60 human graders from diverse backgrounds to participate in multiple phases of assessment for secondary students' short-answer responses. Participants were assigned to one of the three conditions: no AI-powered assessment support; important-word highlights from AI-powered graders as support; and natural-language grading explanations from GenAI-powered graders as support. Mixed-effect regression analyses were adopted to examine the impacts of different AI-powered assessment insights on human assessment practices both in current assessment with AI-powered insights presented and in later assessment without AI-powered insights. Our findings indicate that GenAI-enabled natural-language insights significantly improved educators' feedback quality compared to educators without AI support ($\beta = 0.190$, $p = 0.010$). Important-word highlights from traditional AI graders had negligible impact on educators' feedback quality. Although significant improvements in grading accuracy were not observed, natural-language insights showed greater potential to enhance grading accuracy ($\beta = 0.556$, $p = 0.093$) than important-word highlights ($\beta = -0.060$, $p = 0.848$). Furthermore, educators reported significantly greater satisfaction and willingness

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to adopt the natural-language insights in practice compared to the important-word insights ($p < 0.05$, $r \in [0.352, 0.490]$). Finally, prior exposure to AI-powered insights to some extent fostered educators' more effective assessment practices in subsequent assessment activities without AI support, though further longitudinal research is needed to establish empirical significance.

Keywords Human-AI collaboration · Generative AI · Assessment · Feedback · Grading

1 Introduction

Assessment is fundamental to the educational process, serving as a mechanism that measures student learning progress and guides educators in adapting their instructions (Shepard, 2000; Brown, 2005). As primary facilitators of assessment, educators are expected to leverage assessment to enhance student learning, and promote fairness and accuracy in evaluating student performance (Schafer, 1993; Price, 2012). This requires educators to not only provide reliable and timely grading but also feedback that justifies the grades and offers actionable suggestions for future improvements (Schafer, 1991; Dai et al., 2024). However, educators struggle to deliver timely assessment[1] as evaluating students' responses and providing constructive feedback are typically time-consuming and labour-intensive processes, especially responses in the form of writing (e.g., short-answer questions and essays) (Okoye et al., 2013; Ramesh & Sanampudi, 2022; Henderson et al., 2019; Filighera et al., 2022).

To alleviate such burdens, one approach involves the adoption of Automated Assessment Systems (AAS) that harness the power of AI and natural language processing techniques to automate the scoring and feedback provision of students' written responses (Okoye et al., 2013; Bonthu et al., 2021; Cavalcanti et al., 2021; Dai et al., 2023). However, existing AASs remain prone to assessment errors and oftentimes provide only generic feedback (e.g., a pre-determined sample answer as feedback) (Li et al., 2023; Deeva et al., 2021; Fleckenstein et al., 2023). These issues may adversely affect students' learning outcomes and potentially discourage student engagement (Li et al., 2023; Vetrivel et al., 2024). In addition, to ensure responsible use of AI-based AASs, existing research has emphasised the significance of preserving educators' autonomy in making/validating grading decisions while leveraging AASs (Colonna, 2023; Figueras et al., 2024; Farazouli, 2024; Gnanaprakasam & Lourdusamy, 2024; DiSabito et al., 2025). Therefore, one increasingly explored research initiative focused on enabling the collaboration between educators and AASs in a human-centric manner, where the role of the grading systems is to augment human decision-making instead of replacing human graders (Peeters et al., 2021; DiSabito et al., 2025). Researchers in this area have commonly adopted computational techniques to dissect the decision-making processes behind AASs and then derived insights

[1] In this paper, we use "assessment" interchangeably with "grading" and "feedback provision" as those terms have commonly referred to the "assess" action in literature (Brown et al., 1997; Boud & Molloy, 2013).

from machine assessment to evaluate how such insights impact human assessment (Ramírez et al., 2019; Zeng et al., 2022; Zhao et al., 2023). Researchers, for example, have demonstrated that AI-powered assessment insights, such as important words detected by a machine (Zeng et al., 2022), can be presented to human graders to potentially scale up human assessment performance.

The recent advancements in natural language processing and generative AI (GenAI), e.g., GPT-4, have brought new opportunities to further advance human-AI collaborative assessment (Atchley et al., 2024; DiSabito et al., 2025). GenAI technologies excel at producing contextually relevant explanations in natural language providing the reasons behind their grading decisions (Wu et al., 2024; Xiao et al., 2024). In contrast to assessment insights reported in the existing literature, e.g., machine-derived important words which may impose excessive cognitive efforts to interpret (Zeng et al., 2022), GenAI's natural-language insights offer potentially more intuitive and accessible explanations to benefit human assessment. However, there remains limited investigation into how such natural-language insights enabled by GenAI influence human assessment, hindering the establishment of effective human-AI collaboration mechanisms to scale up educators' assessment practices. Furthermore, investigations into AI-assisted decision-making in education face growing concerns that educators' increasing reliance on automated support can erode their ability to exercise nuanced, empathetic and agentic judgement, reducing their opportunities to develop and refine their professional skills (Selwyn, 2022; Selwyn et al., 2023), especially in practices necessary to maintain assessment standards (Figueras et al., 2024; Selwyn et al., 2023). Therefore, we conducted a study to investigate the impacts of natural-language assessment insights from GenAI graders on educators' assessment practices, and how educators' prior exposure to such insights affected their future assessment practices. Formally, the paper reports on the findings of the study that aimed to answer the following **R**esearch **Q**uestions:

- **RQ1.** To what extent do natural-language assessment insights enabled by GenAI impact educators' grading?
- **RQ2.** To what extent do natural-language assessment insights enabled by GenAI impact educators' feedback provision?
- **RQ3.** To what extent do educators' prior engagement in AI-augmented assessment affect their future assessment without AI support?

2 Literature Review

2.1 Grading in Assessment

Grading represents one of the foundational components of educational assessment, offering a quantitative measure of student progress (Black & Wiliam, 1998; Guskey, 2001). It promotes students' continual engagement in learning and contributes to their future development (Brookhart, 2012; Butler, 1988; Rosenbaum, 2001). Effective grading provides quantitative measure of student progress, wherein educators identify areas of improvement and suggesting future actions (Hattie & Timperley, 2007; Black & Wiliam, 1998; Guskey, 2001). Specifically, when undertaking grading tasks, educators engage in cognitive activities where they make judgements based on features in students' works, such as content accuracy and method appropriateness with respect to task requirements (Orrell, 2007). The intricate nature of grading necessitates extensive time and effort from educators, especially in assessment tasks requiring open-ended responses (e.g., essays and short-answer questions) (Nilson, 2014). Furthermore, educators' grading processes for these open-ended responses are oftentimes characterised by their subjective judgements (e.g., educators' linguistic preferences Padó and Padó, 2022 and students' in-class behaviours Ferman and Fontes, 2022), leading to biased grading outcomes. As a result, automatic assessment systems capable of autonomously scoring students' open-ended responses have been proposed as the solution to alleviate educators' marking workload while improving scoring objectivity and consistency, particularly in courses with high student enrolments and limited resources (Attali & Burstein, 2006; Hahn et al., 2021).

Traditional AASs have frequently employed natural language processing techniques based on machine learning, such as regression (e.g.,Kumar et al., 2019;Woods et al., 2017) and classification (e.g., Ramesh and Sanampudi, 2022; Lu and Cutumisu, 2021), to automate the scoring of student written responses. Some of these systems have achieved decent grading performance. For instance, (Xue et al., 2021) implemented a multi-task learning approach based on Bidirectional Encoder Representation from Transformers (BERT), achieving a performance of 0.830 (measured by Quadratic Weighted Kappa) on essay scoring tasks. (Lun et al., 2020) adopted BERT with data augmentation techniques To enable a performance of 0.823 (measured by F1 score) on short answer scoring tasks. However, the practical integration of these systems in authentic grading practices are oftentimes hindered by the systems' inability to reason for their scoring decisions (Schlippe et al., 2022).

More recently, driven by GenAI's capability to understand natural-language texts and provide their decisions along with justifications in natural language, researchers are increasingly exploring GenAI technologies as the alternatives to carry out automatic grading tasks (Mao et al., 2024; Giannakos et al., 2024; DiSabito et al., 2025). Researchers have also reported satisfactory performance achieved by GenAI with carefully designed prompts and cautiously selected task-specific examples. For instance, the work by Yan et al. (2024) presented accuracy scores above 0.90 achieved by GenAI in marking less cognitively demanding questions (i.e., questions requiring basic comprehension and application of concepts). Henkel et al. (2024) reported GenAI To achieve around 0.70 Cohen's *kappa* in scoring short-answer responses from K-12 education, similar To the human-level agreement of 0.75 κ. Yet, several other studies evaluating GenAI in scoring open-ended textual responses have commonly suggested that "off-the-shelf" GenAI without fine-tuning (i.e., training the models with task-specific datasets) is not yet ready for practical adoption due to the limited accuracy achieved in those scoring contexts (Chamieh et al., 2024; Schneider et al., 2024). These conflicting findings indicate that GenAI can not yet be entrusted to solely undertake scoring for students' open-ended responses, and a partnership between GenAI and educators may be a more plausible way to facilitate grading practices (DiSabito et al., 2025; Xiao et al., 2024; Schneider et al., 2024). For this reason, we explored in this study whether, and if so, to what extent natural-language assessment insights enabled by GenAI could augment educators' grading processes.

2.2 Feedback in Assessment

Apart from grading, feedback is another crucial element of educational assessment needed to promote students' learning (Hattie & Timperley, 2007; Black & Wiliam, 1998; Sadler, 1989). Effective feedback serves as an essential communication tool for bridging the gap between students' current performance and the desired academic outcomes (Dai et al., 2024; Hattie & Timperley, 2007). Hattie and Timperley (2007) described feedback as occurring in three major types (i.e., feeding up, feeding back, and feeding forward) and having four focus levels (i.e., task, process, self-regulatory, and self). However, feedback in this vein typically conveys task-related information to students, with little consideration on how students perceive and integrate the feedback in their future learning (Winstone et al., 2022). To ensure that students can properly interpret the received feedback and enact on the feedback correspondingly, educators and researchers have increasingly advocated the use of learner-centred feedback that focuses more on students' learning processes rather than solely on students' performance (Winstone et al., 2022; Molloy et al., 2020; Henderson et al., 2019). Adding to this line of work, (Ryan et al., 2021) proposed a conceptual model for providing learner-centred feedback in authentic pedagogical environments, advising that feedback should follow three commonly recognised principles: learner agency (i.e., learners should be the primary actor in the feedback process), learner sense-making (i.e., feedback should be designed in a way that facilitates interpretation for learners), and learner impact (i.e., feedback should be actionable).

The provision of constructive feedback in a timely manner presents significant operational challenges within contemporary educational environments, particularly in contexts characterised by large student cohorts (Henderson et al., 2019). Due to these challenges, researchers are increasingly exploring technical solutions to automate the feedback provision for students. For example, *RATsApp*, designed for STEM education, provided students with formative feedback, scaffolding, and real-time performance tracking to enhance key competencies such as mathematical literacy, data literacy, and representational competence (Steinert et al., 2024). In a different vein, *AcaWriter* offered real-time feedback on academic and professional writing, helping users improve their writing skills by identifying key rhetorical elements and offering suggestions for clarity, structure, and coherence (Knight et al., 2020). Majority of the existing AASs facilitate feedback provision for close-ended tasks, where predetermined ground-truth answers are available (Dai et al., 2024). Meanwhile, AASs enabling feedback provision for open-ended tasks frequently rely on the comparisons between student answers and the desired solutions to generate feedback, with little consideration on how students perceive the feedback (Cavalcanti et al., 2021).

To bridge the gap in existing AASs and enable more accessible feedback for open-ended student responses, researchers are increasingly investigating the feasibility of GenAI technologies as a feedback provider. For example, (Dai et al., 2024) reported more readable, consistent and process-focused feedback generated by GPT-4 for students' written project proposal compared to that by educators. The work by (Meyer et al., 2024) suggested that feedback generated by GPT-3.5 increased secondary students' text revisions, motivation and positive emotions. Despite the promises, several critical issues exist which hinder the practicality of GenAI technologies for feedback provision. First, the reliability of the generated feedback remains a concern. For instance, (Jia et al., 2024b) reported that the GenAI-produced feedback contained hallucination (i.e., generating irrelevant, made-up, or inconsistent content). Second, the accuracy of the generated feedback for authentic learning environments remains a work-in-progress. Dai et al. (2024) noted that GPT-4 failed to accurately assess students' performance in the feedback. Similar findings were echoed in pertinent research (Jia et al., 2024a; Gabbay & Cohen, 2024). Third, the acceptance of the GenAI-generated feedback among students remains an open question. Students have commonly expressed a higher preference for feedback formulated by educators instead of by GenAI technologies in several studies (Nazaretsky et al., 2024; Jia et al., 2024a). Given these reported limitations on GenAI-produced feedback, we argue that GenAI should function as a collaborative agent to augment human educators' feedback provision for open-ended student responses rather than serving as a standalone, autonomous feedback system. To this end, we investigated whether assessment insights drawn from GenAI's assessment of students' answers could serve as effective auxiliary information for educators' feedback provision.

2.3 Human-AI Collaboration in Assessment

Within the theoretical framework of human-AI collaboration, there are three commonly recognised modes of collaboration: AI-centric, symbiotic, and human-centric (Fragiadakis et al., 2024). In the case of automated educational assessment, the AI-centric mode refers to the studies discussed in Section 2.1 and Section 2.2, where the AI-powered AASs lead the decision-making processes, finalising the scores and feedback, then deliver these outcomes to educators. The symbiotic mode typically involves the *Learning to Defer* paradigm (Fragiadakis et al., 2024), where confident decisions made by AI-powered AASs are entrusted, while uncertain ones are deferred to human graders for further assessment (Funayama et al., 2020, 2022; Li et al., 2023; Schneider et al., 2023). Yet, both the AI-centric and symbiotic modes result in assessment decisions that heavily rely on the machines' decision-making (i.e., in AI-centric mode, all decisions are made by AI; in symbiotic mode, confident decisions made by AI are retained (Li et al., 2023)).

Assessment decisions from AASs are prone to errors, even in cases where the AASs are confident about these decisions (Galici et al., 2023). Therefore, the reliance on AASs' assessment decisions raises concerns about inaccuracy and the potential adverse effects of incorrect assessment on student learning (Figueras et al., 2024; Yan et al., 2024). Students may also feel discouraged and disengaged upon discovering that their scores or feedback are generated by AASs rather than by teachers (Figueras et al., 2024; Nazaretsky et al., 2024). In addition, to ensure the responsible adoption of AI-powered AASs in education, existing literature has put a strong emphasis on the educator autonomy in making or validating the assessment outcomes, working collaboratively with AASs (Colonna, 2023; Figueras et al., 2024; Farazouli, 2024; Gnanaprakasam & Lourdusamy, 2024; DiSabito et al., 2025). This underscores the need for *human-centric modes of collaboration* in educational assessment, where AI acts as a facilitator to augment human decision-making, allowing educators to retain their autonomy and apply their judgment to deliver high-quality assessments (Fragiadakis et al., 2024; Peeters et al., 2021).

To facilitate human-centric collaborative assessment, various AI-powered insights can be extracted from AASs. These insights include automated scoring decisions, the confidence levels associated with these decisions, the important words that lead to these decisions, and the natural-language explanations of assessment rationales underlying these decisions (Zeng et al., 2022; Xiao et al., 2024; Lee et al., 2024; Jiang & Bosch, 2024; Li et al., 2025). Prior research has demonstrated that leveraging such AI-powered insights can enhance human assessment practices to varying degrees. For instance, (Xiao et al., 2024) reported that GenAI-produced scoring decisions, scoring confidence and natural-language explanations enabled educators to revise their grading decisions, leading to improved assessment consistency among human graders. However, these AI-powered insights may have different implications for educators' assessment practices. Educators, particularly those with less experience, might unintentionally align their evaluations with AI-generated decisions if they are presented, regardless of the correctness of these decisions (Flodén, 2025; Hall et al., 2024; Swiecki et al., 2022). Furthermore, the presence of machine scoring confidence during human assessment could lead to human oversight in assessing student responses confidently graded by machine (Li et al., 2025).

In response to these challenges, researchers have focused on deriving insights that reveal the decision-making processes of AASs as auxiliary information to support human grading (Zeng et al., 2022; Tornqvist et al., 2023; Li et al., 2025). For example, Zeng et al. (2022) demonstrated that presenting machine-identified important words during human assessment showed potential to improve human assessment accuracy. Despite the promises of important-word insights to augment human assessment, researchers have also noted the additional cognitive load[2] they may impose on graders attempting to leverage them (Li et al., 2025). Inspired by the superiority of GenAI in reasoning its decision-making process by generating intuitive natural-language explanations (Wu et al., 2024; Xiao et al., 2024), we were motivated to investigate the role such natural-language insights could play in human assessment.

To date, limited empirical works have investigated whether these natural-language insights enabled by GenAI can benefit educators during assessment to streamline both their grading and feedback provision processes. This gap in the literature presents a critical barrier to establishing robust collaborative mechanisms between educators and GenAI-based ASSs. Moreover, concerns persist regarding the potential for over-reliance on AI-assisted decision-making in education, which may erode educators' professional competencies over time (Selwyn, 2022; Selwyn et al., 2023). Researchers have discussed that the dependency on support offered by AAS may diminish educators' skills essential for maintaining assessment standards (Figueras et al., 2024; Selwyn et al., 2023). It is therefore imperative to foster a human-centric

[2] Cognitive Load Theory is a psychological theory that describes how the human cognitive system processes and manages information, suggesting that individuals' task performance is directly influenced by how cognitive resources are allocated during information processing (Sweller, 1988).

collaborative approach between educators and GenAI in a way that sustainably bolsters educators' assessment practices in grading and feedback provision without sacrificing their professional development. Therefore, in the present study, we examined (1) the effects of GenAI-enabled natural-language assessment insights on educators' current assessment tasks (RQ1 & RQ2) and (2) how educators' prior exposure to these natural-language insights affected their assessment practices in future assessment tasks without such insights (RQ3).

3 Methodology

3.1 Grading Task Description

To address our research questions, we examined the short-answer grading tasks from the *ASAP-SAS* dataset[3]. This dataset has been made publicly available and frequently adopted in recent research exploring the potential of GenAI in assessing student textual responses (e.g.,Xiao et al., 2024;Chamieh et al., 2024; Jiang and Bosch, 2024). Moreover, using this dataset may further promote the generalisability and reproducibility of our findings. Specifically, *ASAP-SAS* included ten question prompts covering subjects Like biology, science and English learning. Responses in English from Grade 10 students to the question prompts, alongside the scores awarded for their responses, were provided. Driven by the prevalent research interest in supporting educators' evaluation of students' language learning and scientific reasoning skills (Hahn et al., 2021), we opted To examine questions 9 (**Q9**) and 10 (**Q10**). **Q9** is designed to assess students' English skills, as it requires them to summarise how a reading passage has been organised and provide supportive details from the passage to justify their arguments. **Q10** tasks students with a scientific reasoning activity, wherein students select an experimental setting and discuss the potential experiment outcomes of the setting using evidence from the presented experiment data. The two question prompts share the same grade scale of 0, 1 and 2. For each question prompt, a training set for implementing the AAS and a testing set for evaluating the performance of the AAS had been made available. More details about the student answer data (e.g., sizes, distributions of grades) and grading tasks (e.g., reading materials, marking rubrics and sample answers) are available in the sections A1 and A6 of our Digital Appendix[4].

3.2 AI-Powered Assessment Insights

To scrutinise how the natural-language assessment insights generated by GenAI impacted educators' assessment practices and answer our RQs, we involved generative pre-trained language models to produce the natural-language insights. We adopted "off-the-shelf" models as they could be readily adapted to diverse assessment

[3] https://www.kaggle.com/c/asap-sas

[4] https://docs.google.com/document/d/e/2PACX-1vRBeKnzP7Th4PRevjDGBgaZkWdiOKezEw1lcOyUMgmS6k3WYPNUl92xeNzRoJkSbnec_JMqayXu55-Q/pub

contexts, offering greater flexibility and practicality (Henkel et al., 2024). To this end, we adopted GPT-4 as the GenAI grader to derive the natural-language assessment insights driven by its superior performance reported in pertinent research as compared to other general-purpose GenAI models (Tang et al., 2024; Ormerod & Kwako, 2024). We utilised the API[5] offered by *OpenAI* for GPT-4. Since an effective prompt (i.e., user input that guides the GenAI models' responses) was central to ensuring the accuracy of GenAI assessment (Henkel et al., 2024), we engineered our prompts following methods reported to be efficacious in previous studies (Henkel et al., 2024; Lee et al., 2024; Wu et al., 2024; Yancey et al., 2023; Cegin et al., 2024; Stahl et al., 2024). Specifically, we adopted a few-shot prompting approach (i.e., prompting with example graded student answers) demonstrated to be effective in pertinent research (Lee et al., 2024; Henkel et al., 2024), and incorporated additional contextual details into the prompt including task-related information (i.e., subject area, student level, reading materials and student task descriptions) and marking rubrics (Lee et al., 2024; Wu et al., 2024). For the few-shot examples, we included one randomly selected graded answer per grade scale inspired by pertinent work (Yancey et al., 2023; Cegin et al., 2024) (for these examples, see A3 in Digital Appendix[6]). The GPT-4 grader was then instructed to first explain how it assessed the student answer in a step-by-step manner (i.e., chain-of-thought) and then assign an integer score of 0, 1, or 2 to determine the quality of the answer (Lee et al., 2024). Subsequently, these natural-language explanations and the assessment score were outputted in a structured JSON format[7], containing "steps" as a list of explanation texts and "final_score" as the integer score Stahl et al., 2024 (for an example response, see A2 in Digital Appendix[6]). Our complete prompt template is as follows, with information in the square brackets in accordance with task materials and student answers:

> *You are a grader that evaluates Grade 10 students' answers to a question prompt. You will be provided information for the question prompt and a student's answer. Your task is to provide the reasoning for evaluating this answer step-by-step. Then, based on your reasoning, you should finalise a mark for the student's answer.*
> *Topic:* `[Topic]`
> *Reading materials:* `[Reading Materials]`
> *Question:* `[Question]`
> *Rubric:* `[Rubric]`
> *Sample answers:* `[Sample Graded Student Answers]`
> *Answer to be graded:* `[Student Answer]`

To get the definite rationale and scoring decisions from the GenAI grader, we explicitly set its temperature to 0 to avoid randomness in the responses. The perfor-

[5] https://openai.com/index/gpt-4/

[6] https://bit.ly/3Zi5QAi

[7] This was implemented by defining a structured schema output when invoking the API to ensure strict adherence, see https://platform.openai.com/docs/guides/structured-outputs for more details.

mance of the GenAI grader was evaluated by three metrics commonly adopted in AAS research (Bonthu et al., 2021), including accuracy, Quadratic Weighted Kappa (QWK), and macro F1 score (see A1 in Digital Appendix[8]).

To fully demonstrate the impact of these natural-language insights in supporting human educators' assessment practices, we incorporated other AI-powered assessment insights, as demonstrated to be somewhat effective in prior research, to serve as a comparative benchmark. Specifically, words that played crucial roles in the decision-making process of BERT-based automatic assessment (as demonstrated in previous work (Zeng et al., 2022) and described in Section 2.3) could be extracted through an XAI approach and highlighted to offer valuable insights that potentially augmented human graders' assessment of student answers, leading to improved assessment accuracy. Therefore, similar to (Zeng et al., 2022), we implemented BERT-based graders to enable important-word assessment insights in comparison with natural-language insights produced by GenAI.

To identify the sets of words playing critical roles during BERT graders' assessment of student answers, we first implemented a BERT-based grader for each of the selected question prompts based on BERT-base-uncased from *HuggingFace*[9] to classify student answers into corresponding grade scales. We carried out five-fold cross validation similar to prior research (Sawatzki et al., 2022) to ensure reliable measurement of BERT-based graders' performance. Specifically, for each question prompt, its corresponding training set was partitioned into five equal-size portions, and an automatic grader was trained with four of the five portions and validated with the remaining one. This process was repeated until each portion had been utilised as the validation set once, leading to five trained BERT graders for each question prompt. We leveraged the testing set to measure the performance of the five graders and reported their average performance for each question prompt by the same metrics as used to evaluate the GenAI graders (see A1 in Digital Appendix[8] for details on grader performance). The five graders for each question prompt were retained for obtaining important-word assessment insights. This process typically involves the adoption of XAI techniques (Zeng et al., 2022; Tornqvist et al., 2023; Poulton & Eliens, 2021), which quantitively measures the significance of individual words that led to BERT graders' assessment decisions. In our study, we adopted *integrated gradients* as it had been validated in previous research to produce reliable and consistent word-level importance (Poulton & Eliens, 2021; Sato et al., 2022). Such importance, referred to as attribution scores (Sundararajan et al., 2017), could be either positive, indicating the significance of specific words leading to the predicted class labels, or negative, suggesting the reduced likelihood of the predicted outputs caused by the words. As the five graders from the cross-validation of each question prompt were fine-tuned with different training data, there could be slight difference in the extracted word-level importance (Poulton & Eliens, 2021). Therefore, when quantifying the significance of each word from the input answers, we averaged the attribution scores of that word across the five graders.

[8] https://bit.ly/3Zi5QAi

[9] https://huggingface.co/bert-base-uncased

While these numeric attribution scores could be contrasted to identify words that are more important than other words in a student answer, there was no common approach to establish a significance threshold that characterised words as important/unimportant (Bolotova et al., 2020). To determine the words that were more important within individual answers, researchers have primarily utilised the approach that relied on human graders' manual annotation of words which signalled answer qualities (Zeng et al., 2022; Bolotova et al., 2020). Specifically, human graders were invited to annotate words that led to either the attainment or the deduction of marks. Subsequently, the number of annotated words and the total number of words within the answers were utilised to implement a linear regression model that estimated the n number of important words given the Total number of words within an answer. Akin To this approach, we invited two human graders to independently annotate words reflecting answer qualities for the 72 student answers (i.e., 36 answers for each question prompt) adopted in our human study discussed later in Section 3.3. Thereafter, we implemented the linear regression model to determine the n for each student answer, which yielded a Linear coefficient of 0.332, i.e., for an answer with 10 words, approximately 3 words would be important for determining answer quality (see A5 in Digital Appendix[10] for detailed statistical results). We then sorted the word-level importance according to its absolute attribution scores to account for words both contributing to and working against assessment decisions made by BERT graders, as both positively and negatively attributed words essentially affected machine's decision-making (Sundararajan et al., 2017). The top n most important words within each answer according to their sorted importance were leveraged as the important-word assessment insights from BERT graders.

3.3 Human Study Design

3.3.1 Participants

Previous research has documented that crowd-sourced graders, particularly those with relevant prior experience, provide high-quality assessments of student written responses, comparable to that of expert graders (Ahn et al., 2021). Inspired by this, we recruited participants from *Prolific*[11], a crowd-sourcing platform that connects researchers with a diverse, global pool of participants for conducting studies and experiments. To ensure that only participants with relevant expertise were involved, we applied a pre-screening filter, allowing only participation from graders with teaching experience in relevant subjects (i.e., English and Science) currently employed at educational institutions. Furthermore, given that the assessment tasks were in English, the graders were also expected to have English as their "first language" or "fluent language".

We recruited a Total of 60 participants from 7 different regions[12], including Asia, Australia and New Zealand, Europe, Middle East and North Africa, North America (United States, Canada, Mexico), South America and Sub-Saharan Africa. Among the 60 participants, 47 identified as female and 13 as male. The majority of the partic-

[10] https://bit.ly/3Zi5QAi

[11] https://www.prolific.com/

[12] Ethics approval has been obtained from Monash University (Project ID 30074).

ipants reported having completed either a four-year undergraduate degree or a professional degree. A Total of 59 participants indicated either fluent or native proficiency in English, while one reported conversational-level proficiency. Regarding teaching experience, 13 participants reported having 1-3 years of experience, 11 reported 3-6 years, 8 reported 6-9 years and 28 indicated over 9 years of teaching experience. In terms of current professional roles, 53 participants were K-12 educators, 6 worked in early childhood education, and 1 was employed in higher education.

The 60 participants were randomly and evenly assigned to assess student responses from one of the two selected question prompts. The experimental data (e.g., participants' assigned scores and feedback for students' answers) were collected via a survey in *Qualtrics*. To reduce bias in the collected data (e.g., different grading time caused by variations in screen sizes), all participants were required to start the study using personal computers. Furthermore, as crowd-sourced participants' demographics and behaviours have been reported to likely vary over time (Stewart et al., 2017), we carried out our study using an iterative approach to avoid variance caused by participants' varying backgrounds. Specifically, we completed the data collection in a batch-by-batch manner, where we divided the 60 participants into 10 batches and collected each batch at different times of the day. In addition, we explicitly declared that participants should not exploit other technologies during their assessment of student answers to avoid contamination of the collected data. Information regarding the study goals and procedures was clearly communicated to the participants in the survey introduction. Subsequently, they were asked to provide their consent and demographic information prior to the commencement of the study. As four participants did not complete the study entirely, their data were excluded from our analyses.

3.3.2 Study Phases

Our designed study consisted of four phases as illustrated in Fig. 1, and as described below.

Pre-Training The Pre-Training phase aimed for participants to acquaint with the materials related to the assessment tasks. In this phase, participants were first asked to read through these materials, including the reading content (i.e., the reading passage for **Q9** and the science experiment details for **Q10**), the question prompts to which students responded, and the marking rubrics. Then, the participants were provided two sample answers per grade scale to get familiar with the formats of students' answers. Lastly, we presented a set of recommended guidelines for feedback provision, accompanied by examples corresponding to each guideline (see A4 in Digital Appendix[13] for the complete guidelines). These guidelines were drawn from the learner-centred feedback framework (Ryan et al., 2021), which had been recognised as an effective feedback practice across different educational levels (Winstone et al., 2022; Molloy et al., 2020; To et al., 2023). To ensure participants' engagement in reading the materials and filter out potential uncommitted participants, research conducting online surveys typically incorporated attention check questions (Hauser &

[13] https://bit.ly/3Zi5QAi

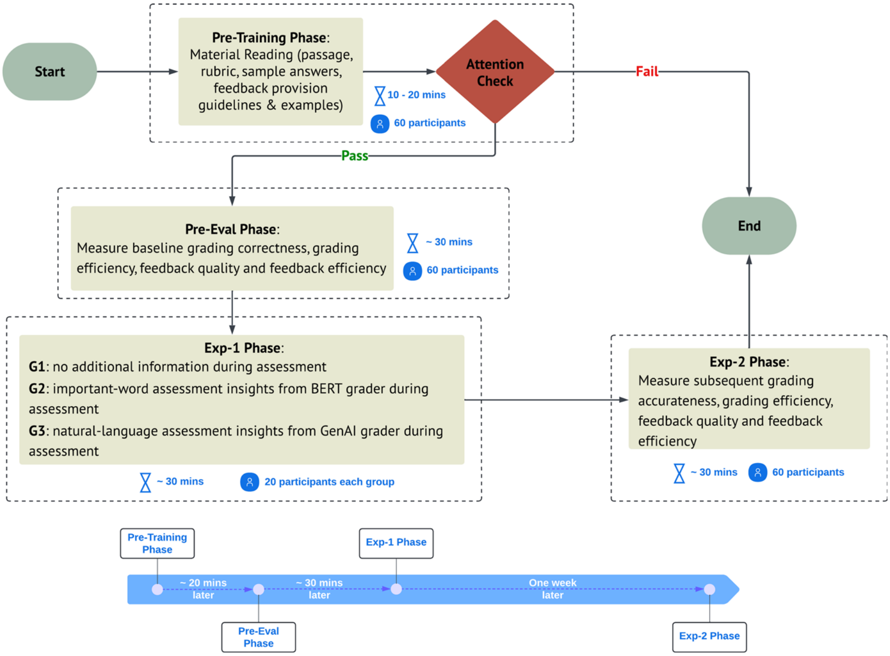


Fig. 1 The flow chart describing study procedures

Schwarz, 2016; Abbey & Meloy, 2017; Kung et al., 2018). Likewise, we introduced two attention check questions related to the reading materials (see A7 in Digital Appendix[13]). Participants should at least answer one of the two questions correctly in order to qualify for the subsequent phase.

Pre-Eval The purpose of the Pre-Eval phase was to measure participants' baseline performance in grading and feedback provision before being presented AI-powered assessment insights. To this end, we randomly sampled 12 student answers for each question prompt, with 4 from each grade level. Participants were asked to assess the answers for their assigned question. Specifically, they first visited a scoring page to determine a grade for an answer, then navigated to a feedback page to provide feedback correspondingly. During the assessment processes, they could access the information available in the Pre-Training phase from the left panel if needed (see Fig. 2). Figure 3 presented the right-side panel layouts of the scoring page and the feedback page, respectively. Upon completion of the Pre-Eval phase, the participants moved on to the following phase.

Exp-1 The Exp-1 phase was designed to answer our RQ1 and RQ2 regarding the extent of effects from GenAI natural-language assessment insights on educators' assessment process. Correspondingly, we established three groups, denoted as **G1**, **G2** and **G3** to contrast the effects of the natural-language insights with no addi-

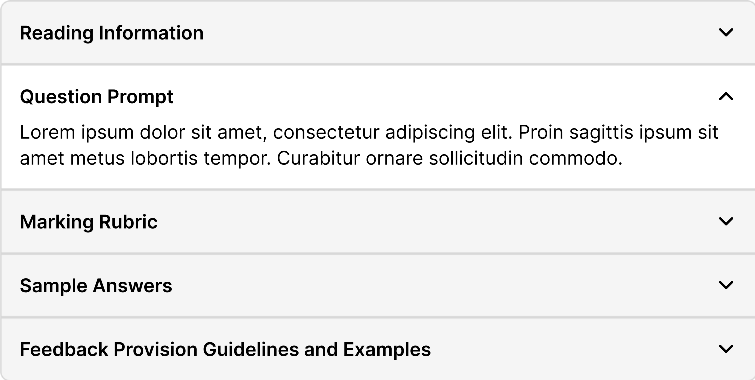
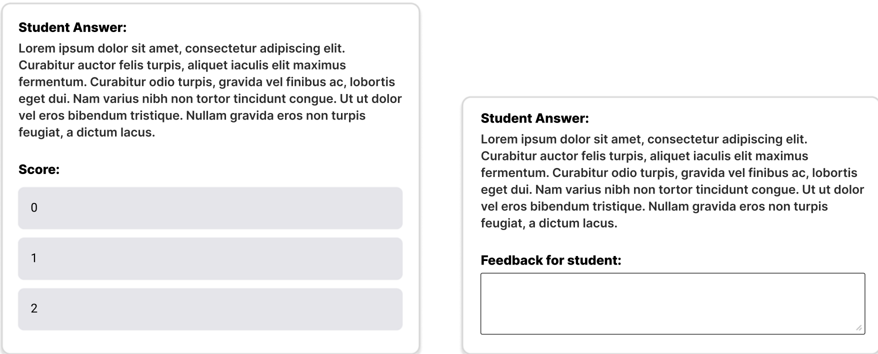


Fig. 2 Layout of the left-side information panel, where graders can click on an item to toggle and read. *Note: Placeholder text is used here to demonstrate the layout, which do not convey any actual meaning. In actual experiments, the placeholder text is replaced correspondingly by task-related materials (e.g., question prompt, marking rubric) in English*



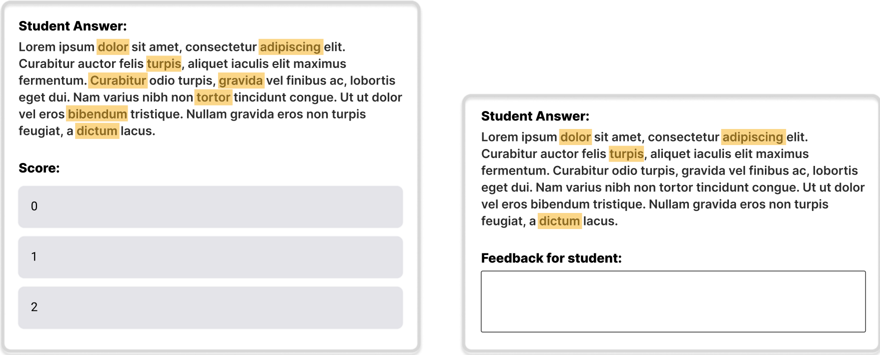
(a) Scoring Panel (b) Feedback Panel

Fig. 3 Layouts of the right-side scoring and feedback panels in the Pre-Eval phase. *Note: Placeholder text is used here to demonstrate the layout, which do not convey any actual meaning. In actual experiments, the placeholder text is replaced correspondingly by actual student responses in English*

tional insights and important-word insights from BERT (as detailed in Section 3.2), respectively.

G1 Participants were to assess the answers without any AI-powered insights.
G2 Participants were to assess the answers with important-word insights from BERT graders.
G3 Participants were to assess the answers with natural-language insights from GenAI graders.

The 60 participants (30 per question prompt) were randomly and evenly assigned To these three groups, with each group containing 20 graders – 10 drawn from each question prompt. We randomly sampled 12 unseen answers (4 from each grade



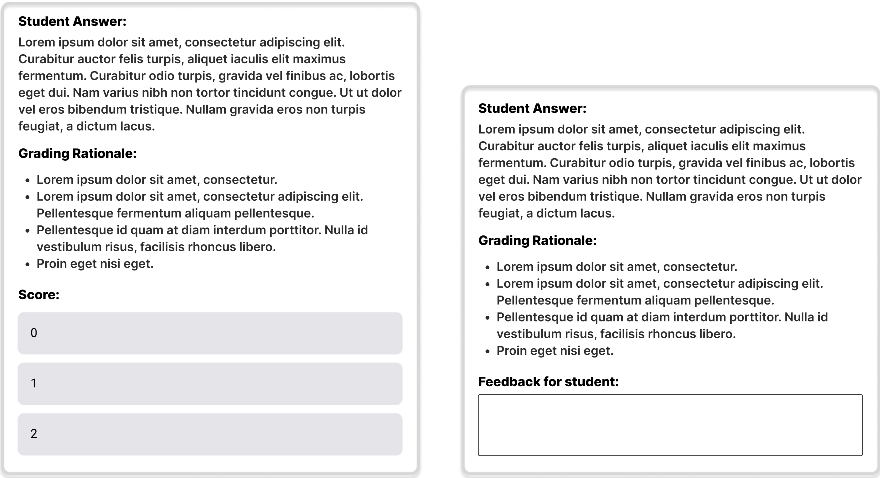
(a) Scoring Page (b) Feedback Page

Fig. 4 Layouts of the right-side scoring and feedback panels for G2, where the highlighted words were deemed important in BERT graders' assessment process. *Note: Placeholder text is used here to demonstrate the layout, which do not convey any actual meaning. In actual experiments, the placeholder text is replaced correspondingly by actual student responses in English*

level) for each question prompt, and requested participants to assess these answers. It is worth noting that, to mitigate variance caused by inconsistent grading decisions between BERT and GenAI graders, which potentially invalidated our comparisons, we opted for answers correctly assessed by both types of graders, and left the investigation of incorrect ones to our future work. During assessment of the answers, participants from all groups could access all the information available in the Pre-Training phase from the left panel (see Fig. 2). However, the information provided on the right panel (i.e., whether having access to AI-powered assessment insights and the types of insights) was in accordance with participants' allocated group. Participants who were allocated to **G1** utilised the same assessment layouts as in the Pre-Eval phase (see Fig. 3). For **G2** participants, the important-word insights enabled by BERT graders were presented in the form of important word highlighting akin to previous research (Zeng et al., 2022) (see Fig. 4). Participants from **G3** received the natural-language insights from GenAI as bullet-point items as these insights were the step-by-step explanations of GenAI's assessment rationales (see A8 in Digital Appendix[14] for examples of GenAI assessment rationales). Figure 5 illustrated the assessment layouts for G3.

Throughout the entire phase, participants' assigned grades and provided feedback for the answers and their time spent were recorded. Upon completion of all assessment tasks for this phase, participants from **G2** and **G3** engaged in several survey questions collecting their subjective perceptions regarding the AI-powered assessment insights presented during assessment using the five-point Likert-scale from "Strongly Disagree" to "Strongly Agree". Specifically, inspired by previous research (Schlippe et al., 2022; Hoffman et al., 2023; Xiao et al., 2024), we gathered their

[14] https://bit.ly/3Zi5QAi



(a) Scoring Page (b) Feedback Page

Fig. 5 Layouts of the right-side scoring and feedback panels for G3, where the bullet-items explained the assessment process by GenAI graders. *Note: Placeholder text is used here to demonstrate the layout, which do not convey any actual meaning. In actual experiments, the placeholder text is replaced correspondingly by actual student responses and GenAI-produced grading explanations in English.*

opinions on the following questions (note that the same questions were asked respectively for grading and feedback provision):

Q1 I find the AI-powered insights informative and useful for my grading / feedback provision.

Q2 I find the AI-powered insights easily comprehensible for my grading / feedback provision.

Q3 The AI-powered insights help me finalise the grades / compile feedback for the answers more efficiently.

Q4 I would be more willing to grade answers / compile feedback with the AI-powered insights.

Exp-2 To answer RQ3, we further designed the Exp-2 phase which took place one week after the Exp-1 phase. The same set of 60 participants were invited back To score and provide feedback for another set of 12 unseen answers (4 from each grade level) from their previously allocated question prompt. Participants were only provided with information available from the Pre-Training phase during assessment, i.e., there was no AI-powered insights provided to the participants throughout the whole experimental phase. Similarly, throughout this phase, we collected their provided scores and feedback, as well as their time spent engaging in the assessment activities. Our assumption regarding participants' assessment performance was that, if educators from **G2** and **G3** learned from the AI-powered insights from the Exp-1 phase, they would exhibit better assessment performance compared to educators from **G1**, who did not receive such insights.

3.3.3 Data Analysis

With 56 participants completing our study, our final dataset included assessment outcomes by 19 participants from **G1**, 19 participants from **G2**, and 18 participants from **G3**, Totalling to 2,016 instances (i.e., 36 assessed answers for each participant).

For scores that participants assigned, we determined their correctness by comparing to the ground-truth scores students actually received for their answers, producing binary outcomes of "correct" and "incorrect". In terms of the feedback quality, two coders with expertise in learner-centred feedback engaged in the evaluation of the collected feedback. Such evaluation was carried out according to the three key elements of feedback (i.e., learning impact, sense-making and agency), aligning with common conceptualisations of learner-centred feedback in existing literature (Ryan et al., 2021). Each dimension was annotated in a binary manner, reflecting whether said dimension was present in the feedback. The two coders reached an agreement of 0.698 measured by Cohen's κ, indicating substantial agreement. Upon further investigation into the inconsistencies, most conflicts stemmed from ambiguities regarding whether certain sentences in positive tones within the feedback should be categorised as "agency", "sense-making" or both. These conflicts were resolved in a case-by-case manner between the two coders. Subsequently, we determined the quality of each feedback by counting the number of feedback dimensions presented.

In our analysis, our overarching goal was to examine the effects of natural-language assessment insights from GenAI on human assessment performance. While it was reasonable to anticipate that participants' continuous engagement in assessment tasks would lead to performance improvements from Pre-Eval to Exp-1 for all three groups (i.e., practice effects (McCabe et al., 2011)), we posited that the provision of AI-powered insights would have additional positive impacts on educators' assessment performance. Specifically, we formulated the following hypotheses:

- **H1** If the AI-powered insights indeed benefited human assessment (RQ1 & RQ2), participants from **G2** and **G3**, who were exposed to these insights, would exhibit greater performance improvements from Pre-Eval to Exp-1 compared to those from **G1**;
- **H2** If the natural-language insights presented in **G3** were more effective than the important-word insights presented in **G2** (RQ1 & RQ2), **G3** participants would yield more performance improvements than **G2** participants;
- **H3** If prior exposure to AI-powered insights facilitated learning from educators (RQ3), participants in these conditions (i.e., **G2** and **G3**) would establish more effective assessment practices, leading to greater magnitude of performance improvement between Pre-Eval and Exp-2 phases compared to those without prior exposure to such insights (i.e., **G1**).

To rigorously evaluate these hypotheses, we adopted mixed-effect regression analysis to measure the impacts of AI-powered insights on assessment performance. Such a statistical approach is well-suited for our analysis as it accounts for both fixed effects and random effects (Gelman & Hill, 2006). Specifically, we aimed to directly estimate the assessment performance differences attributable to the fixed effects, including the groups to which participants were assigned, the phases of the assessment being recorded, and the interactions between these two factors. More importantly, as assessment performance could be affected by individual human graders' attributes (e.g., their linguistic preference) and the characteristics of student responses (e.g., answer commonness) Padó and Padó, 2022, we sought to account for variance caused by such random factors when scrutinising the differences by introducing these factors as random effects in our analyses. We utilised the lmer4 R package version 3.1-3 to facilitate the implementation of the mixed-effect regression models.

For RQ1, we examined two primary metrics in our analyses: **grading correctness** (i.e., whether or not an assigned score aligned with the score received by the student) and **grading duration** (i.e., the duration from the time an answer was presented till the time a grading decision was finalised). To evaluate grading correctness, we employed a mixed-effect logistic regression model. This approach was suitable for analysing binomial outcomes, allowing us to estimate the likelihood of a correct grading decision for each instance (see Eq. 1). Specifically, the model predicted the grading correctness ($Y_{\text{Correctness}}$) by considering several key variables: β_0 served as the intercept term, representing a baseline condition; β_1 was the coefficient of Phase, explaining how the grading correctness varied when switching from one phase to another while the other variables were kept constant; β_2 was the coefficient of Group, quantifying the difference in grading correctness among the groups in a baseline phase; β_3 was the coefficient of the interaction effect between Phase and Group, capturing the differential impact of experimental phases across distinct participant groups; and $\gamma_{\text{Participant\_ID}}$ and $\gamma_{\text{Answer\_ID}}$ accounted for individual variations among participants and answer characteristics, respectively. For grading duration, we utilised a mixed-effect linear regression model, focusing on the continuous outcomes of the time spent grading (see Eq. 2; the error term ϵ explicitly represented unexplained variability in a continuous outcome in linear regression).

$$\begin{aligned} Y_{\text{Correctness}} = &\ \beta_0 + \beta_1 \times \text{Phase} + \beta_2 \times \text{Group} + \beta_3 \times (\text{Phase} \times \text{Group}) \\ &+ \gamma_{\text{Participant\_ID}} + \gamma_{\text{Answer\_ID}} \end{aligned} \tag{1}$$

$$\begin{aligned} Y_{\text{Duration}} = &\ \beta_0 + \beta_1 \times \text{Phase} + \beta_2 \times \text{Group} + \beta_3 \times (\text{Phase} \times \text{Group}) \\ &+ \gamma_{\text{Participant\_ID}} + \gamma_{\text{Answer\_ID}} + \epsilon \end{aligned} \tag{2}$$

Apart from these objective measures, we analysed the subjective ratings for the survey questions asked at the end of the Exp-1 phase from participants in **G2** and **G3** to understand how they perceived the AI-powered assessment insights in their grading process. Specifically, we ran Mann-Whitney U tests[15] To statistically compare: 1) the ratings of each group to the neutral level to understand the group's overall attitudes; and 2) the ratings between the two groups to understand which AI-powered insights were perceived more positively.

For RQ2, we analysed the data in terms of **feedback quality** (i.e., the number of presented feedback dimensions according to learner-centred feedback) and **feedback**

[15] For effect size, we adopted the formula: $r = \frac{|Z|}{\sqrt{N}}$

duration (i.e., the duration from when an answer was presented to when the feedback was finalised). We implemented a mixed-effect linear regression model to analyse the feedback quality as a continuous outcome (see Eq. 3). For feedback duration, we utilised a mixed-effect linear regression model to examine the time participants spent providing feedback (see Eq. 4). It is worth noting that, for these analyses, we included an additional variable of participants' assigned scores. This inclusion was based on the rationales that educators were likely to provide longer and more comprehensive feedback for lower-quality answers to foster students' understanding and enactment of the feedback (Wu & Schunn, 2020), leading to the longer duration compiling the feedback. However, since the correlations between assigned scores and feedback quality / duration were not our main point of analyses, the variable for assigned scores was introduced as a random effect to account for the variability caused by the assigned scores. Furthermore, similar to RQ1, we looked into participants' subjective responses to the survey questions asked at the end of the Exp-1 phase regarding their perceptions of the assessment insights on their feedback provision.

$$\begin{aligned} Y_{\text{Quality}} = \beta_0 + \beta_1 \times \text{Phase} + \beta_2 \times \text{Group} + \beta_3 \times (\text{Phase} \times \text{Group}) \\ + \gamma_{\text{Participant\_ID}} + \gamma_{\text{Answer\_ID}} + \gamma_{\text{Assigned\_Score}} + \epsilon \end{aligned} \tag{3}$$

$$\begin{aligned} Y_{\text{Efficiency}} = \beta_0 + \beta_1 \times \text{Phase} + \beta_2 \times \text{Group} + \beta_3 \times (\text{Phase} \times \text{Group}) \\ + \gamma_{\text{Participant\_ID}} + \gamma_{\text{Answer\_ID}} + \gamma_{\text{Assigned\_Score}} + \epsilon \end{aligned} \tag{4}$$

For RQ3, we contrasted educators' performance in grading and feedback provision between the Pre-Eval and Exp-2 phases to measure how their assessment practices differed after going through the Exp-1 phase with different assessment settings. The same mixed-effect regression equations for RQ1 and RQ2 were adopted, with assessment data collected for the Pre-Eval and Exp-2 phases, to implement these statistical models.

4 Results

4.1 RQ1: Impacts of GenAI Natural-Language Assessment Insights on Grading

Table 1 details the results from our mixed-effect logistic regression analysis for participants' grading correctness. The Intercept refers to a baseline case (i.e., Phase="Pre-Eval", Group="G1"), suggesting that participants from **G1** had a 0.380 log-odd to correctly grade an answer in Pre-Eval phase. In the Pre-Eval phase, the **G2** participants were slightly more likely to score answers correctly compared to those from **G1** ($\beta = 0.126$, $p = 0.598$; an increase of log-odds by 0.126 relative To the baseline of 0.380), while **G3** participants' log-odds for correctly scoring answers were lower than those of **G1** ($\beta = -0.075$, $p = 0.756$). However, none of these differences were statistically significant, suggesting that the three groups of participants shared similar initial grading accuracy. When moving on from the Pre-Eval phase to the Exp-1 phase, **G1** participants had a statistically significant increase in their log-odds for scoring answers correctly ($\beta = 1.082$, $p < 0.001$; an improved log-odds by 1.082

Table 1 The mixed-effect logistic regression results for **grading correctness** and the mixed-effect linear regression results for **grading duration** between the Pre-Eval and Exp-1 phases

	Grading Correctness				Grading Duration			
Predictors	β	**SE**	**z value**	**p-value**	β	**SE**	**t value**	**p-value**
Intercept	0.380	0.232	1.638	0.101	38.363	4.675	8.207	$< 0.001^{***}$
Group								
G2	0.126	0.239	0.528	0.598	-7.892	5.395	-1.463	0.147
G3	-0.075	0.241	-0.311	0.756	-7.280	5.470	-1.331	0.187
Phase								
Exp-1	1.082	0.222	4.870	$< 0.001^{***}$	-12.070	3.465	-3.483	$< 0.001^{***}$
Interactions								
Exp-1 × G2	-0.060	0.315	-0.192	0.848	2.364	4.901	0.482	0.630
Exp-1 × G3	0.556	0.331	1.681	0.093	8.159	4.968	1.642	0.101

Note: SE – Standard Error; *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

relative To the baseline of 0.380). The **G2** participants, who received important-word insights from BERT graders, exhibited similar but slightly lower likelihood of scoring answers correctly compared to participants from **G1** ($\beta = -0.060$, $p = 0.848$; a slight reduction of log-odds by 0.060 relative to **G1**'s changes of 1.082). Meanwhile, participants from **G3**, who were presented with natural-language insights from GenAI graders, demonstrated the highest magnitude of improvement in likelihood for grading answers correctly relative to those from **G1** ($\beta = 0.556$, $p = 0.093$). Furthermore, the grading accuracy among participants from the three groups (see Fig. 6) revealed that majority of participants produced highly accurate grading outcomes in the Exp-1 phase, even in the absence of auxiliary AI-powered insights (**G1**). This suggested that the grading tasks were relatively straightforward and easy-to-grade for the recruited educators, and hence, the potential for the AI-powered insights to significantly boost educators' grading correctness appeared to be limited.

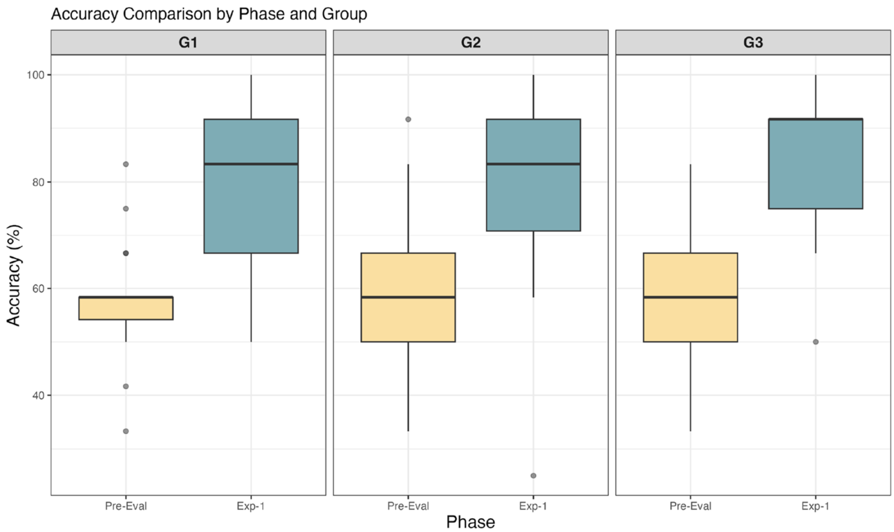


Fig. 6 Distributions of Grading Accuracy among **G1**, **G2** and **G3**

Our analysis for grading duration from Table 1 indicated that participants from **G1** generally Took 38.363 seconds to score an answer in the Pre-Eval phase. Meanwhile, participants from both **G2** and **G3** spent approximately 7.5 seconds less to score answers compared to **G1**, but the differences were not statistically significant. In general, **G1** participants' grading duration reduced significantly by 12 seconds in the Exp-1 phase relative to the Pre-Eval phase ($\beta = -12.070$, $p < 0.001$). Among the three groups, **G1**, without exposure to additional information, achieved the greatest reduction in grading duration, while the decline for **G3** was the smallest ($\beta = 8.159$, $p = 0.101$; around 8 seconds more than the 12-second reduction from **G1**). However, there was no statistical evidence to suggest the disparity in grading duration reduction across the three groups.

Figure 7 summarises the ratings provided by **G2** and **G3** participants regarding their perceptions for the AI-powered assessment insights on grading. The **G2** participants in general considered the important-word insights provided by BERT graders informative and useful (Q1: $M = 3.474$, $SD = 1.349$; the ratings were significantly higher than the neutral level, $p = 0.019$, $r = 0.315$) for grading. However, these AI-powered insights were deemed to be not necessarily easily comprehensible (Q2: $M = 3.211$, $SD = 1.475$; insignificant to neutral level, $p = 0.070$) or speeding up the decision-making process during grading (Q3: $M = 2.947$, $SD = 1.471$; insignificant to neutral level, $p = 0.278$). Participants from **G2** overall held neutral attitude towards adopting such important-word insights in their grading practices (Q4: $M = 3.105$, $SD = 1.197$; insignificant to neutral level, $p = 0.265$). In contrast, participants from **G3** were much more positive about the GenAI-produced natural-language insights. They generally agreed that these insights were informative and useful (Q1: $M = 4.389$, $SD = 0.979$), easily comprehensible (Q2: $M = 4.389$, $SD = 0.979$), and scaling up grading efficiency (Q3: $M = 4.222$, $SD = 1.263$), driving their willingness (Q4: $M = 4.222$, $SD = 1.263$) to adopt them in their grading practices. Their provided ratings were significantly higher than both the neutral level ($p < 0.05$, $r \in [0.569, 0.664]$) and the ratings by **G2** participants ($p < 0.05$, $r \in [0.382, 0.472]$).

The results from our RQ1 analyses suggested that the natural-language assessment insights enabled by GenAI were of higher potential to promote educators' grading correctness compared to cases where educators received no support or received important-word insights from BERT graders. On the other hand, the natural-language

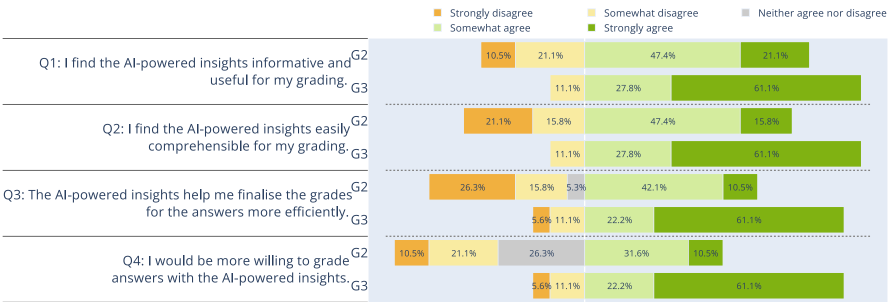


Fig. 7 The subjective ratings for the AI-powered assessment insights by **G2** and **G3** participants for grading

insights could not quantitatively reduce educators' grading time as much as conditions without additional support or with important-word insights. Apart from these objective metrics, educators in general perceived the natural-language insights as a valuable auxiliary source of supporting information during their grading process.

4.2 RQ2: Impacts of GenAI Natural-Language Assessment Insights on Feedback Provision

Looking into the quality of participants' provided feedback (see Table 2), an overall feedback score of 1.8 was achieved by **G1** participants during the Pre-Eval phase ($\beta = 1.800, p < 0.001$). While participants from **G2** and **G3** provided slightly lower quality feedback in the Pre-Eval phase compared to **G1**, their differences in feedback quality were not statistically significant. Moving on to the Exp-1 phase, a slight and insignificant reduction in feedback quality relative to the Pre-Eval phase was observed for **G1** ($\beta = -0.100$, $p = 0.062$). Similarly, participants from **G2** exhibited a negligible reduction in feedback quality ($\beta = 0.065$, $p = 0.366$; the changes in feedback quality for **G2** was 0.065 higher than the -0.100 from **G1**). However, the observed decline in feedback quality for **G1** and **G2** did not extend to participants in **G3**, as the **G3** participants in general provided feedback with 0.090 higher feedback quality (the changes in feedback quality were significantly different between **G1** and **G3**; $\beta = 0.190$, $p = 0.010$).

With respect to feedback duration (see Table 2), **G1** participants spent approximately 118 seconds to provide feedback during the Pre-Eval phase ($\beta = 118.383$, $p < 0.001$), while those from **G2** ($\beta = -42.411$, $p = 0.017$) and **G3** ($\beta = -32.635$, $p = 0.066$) contributed less time compiling feedback. During participants' engagement in the Exp-1 phase, we observed that the duration reduction was most pronounced for **G1** participants ($\beta = -40.205$, $p < 0.001$; a reduction of approx. 40 seconds), followed by **G2** ($\beta = 15.180$, $p 0.056$; a reduction of approx. 25 seconds). **G3** participants achieved the least reduction in feedback duration, which was significantly lower than that of **G1** ($\beta = 20.280$, $p = 0.012$; a reduction of approx. 20 seconds).

Figure 8 summarises the ratings provided by **G2** and **G3** participants regarding their perceptions for the AI-powered assessment insights on feedback provision.

Table 2 The mixed-effect linear regression results for **feedback quality** and **feedback duration** between the Pre-Eval and Exp-1 phases.

	Feedback Quality				Feedback Duration			
Predictors	β	**SE**	**t value**	**p-value**	β	**SE**	**t value**	**p-value**
Intercept	1.800	0.125	14.445	< 0.001***	118.383	13.081	9.050	< 0.001***
Group								
G2	-0.240	0.153	-1.564	0.123	-42.411	17.213	-2.464	0.017*
G3	-0.141	0.155	-0.906	0.368	-32.635	17.447	-1.870	0.066
Phase								
Exp-1	-0.100	0.051	-1.866	0.062	-40.205	5.617	-7.158	< 0.001***
Interactions								
Exp-1 × G2	0.065	0.072	0.904	0.366	15.180	7.918	1.917	0.056
Exp-1 × G3	0.190	0.073	2.596	0.010**	20.280	8.023	2.528	0.012*

Note: SE – Standard Error; *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

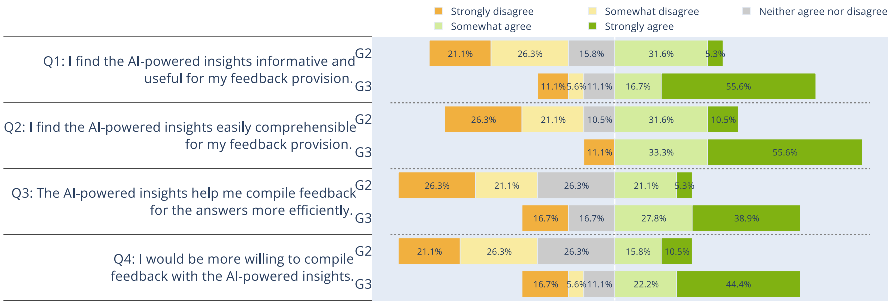


Fig. 8 The subjective ratings for the AI-powered assessment insights by **G2** and **G3** participants for feedback provision

Overall, the **G2** participants were slightly negative about the BERT-enabled important-word insights on their feedback provision. Their overall ratings for informativeness and usefulness ($M = 2.737$, $SD = 1.284$) and comprehensibility ($M = 2.789$, $SD = 1.437$) were lower but not significantly different from the neutral level. They did not consider these AI-powered insights to scale up their feedback efficiency ($M = 2.579$, $SD = 1.261$), nor were they willing ($M = 2.684$, $SD = 1.293$) to adopt them in their feedback provision practices. On the contrary, participants from **G3** generally rated the GenAI-produced natural-language insights positively. They perceived these insights to be informative and useful ($M = 4.000$, $SD = 1.414$), easily comprehensible ($M = 4.222$, $SD = 1.263$), speeding up their feedback provision ($M = 3.722$, $SD = 1.447$), and were hence more willing ($M = 3.722$, $SD = 1.526$) to utilise them to assist in providing feedback. All these dimensions were significantly higher than both the ratings by **G2** participants ($p < 0.05$, $r \in [0.380, 0.664]$) and the neutral level ($p < 0.05$, $r \in [0.352, 0.490]$).

Our findings for RQ2 indicated that the natural-language assessment insights from GenAI enabled educators to improve their feedback quality, while educators with no support or with important-word insights from BERT provided feedback of similar or even lower quality as they continued to engage in the assessment task. While the natural-language insights did not reduce educators' feedback duration as much as the other two conditions, educators generally held more positive attitudes towards such insights in promoting their feedback provision practices.

4.3 RQ3: Impacts of AI-Powered Assessment Insights on Future Assessment Tasks

Table 3 shows that participants' likelihood of grading correctly in the Exp-2 phase increased compared to the Pre-Eval phase, with **G1** participants achieving an overall log-odd increase of 0.258 ($p = 0.196$). The improvement in grading correctness for **G2** was slightly higher than that of **G1** ($\beta = 0.113$, $p = 0.690$), while **G3** achieved the highest magnitude of improvement ($\beta = 0.252$, $p = 0.377$). However, there were no significant differences in the slopes of increase among groups.

Regarding grading duration, Table 3 shows that both **G1** ($\beta = -5.348$, $p = 0.258$; a reduction of approx. 5 seconds) and **G3** ($\beta = 4.206$, $p = 0.535$; a reduction of approx. 1 second) participants exhibited insignificant reduction in grading

Table 3 The mixed-effect logistic regression results for **grading correctness** and the mixed-effect linear regression results for **grading duration** between Pre-Eval and Exp-2 phases

	Grading Correctness				Grading Duration			
Predictors	β	**SE**	**z value**	**p-value**	β	**SE**	**t value**	**p-value**
Intercept	0.330	0.183	1.804	0.071	38.310	5.902	6.491	< 0.001***
Group								
G2	0.117	0.201	0.582	0.560	-7.892	7.488	-1.054	0.295
G3	-0.053	0.202	-0.261	0.794	-7.227	7.592	-0.952	0.344
Phase								
Exp-2	0.258	0.199	1.294	0.196	-5.348	4.723	-1.132	0.258
Interactions								
Exp-2 × G2	0.113	0.285	0.399	0.690	16.323	6.680	2.444	0.015*
Exp-2 × G3	0.252	0.287	0.878	0.380	4.206	6.772	0.621	0.535

Note: SE – Standard Error; *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

duration in the Exp-2 phase relative to the Pre-Eval phase. In contrast, the changes in grading duration for **G2** participants were significantly different from **G1**, with an increase of approximately 11 seconds in the Exp-2 phase compared to the Pre-Eval phase ($\beta = 16.323$, $p = 0.015$).

In terms of feedback quality (refer to Table 4), participants from all three groups provided feedback of comparable quality in the Exp-2 phase relative to the Pre-Eval phase. Among the three groups, **G1** exhibited a slight and insignificant reduction in feedback quality ($\beta = -0.078$, $p = 0.114$), while both **G2** ($\beta = 0.087$, $p = 0.213$) and **G3** ($\beta = 0.105$, $p = 0.138$) produced better quality feedback. However, the differences in the changes of feedback quality among the groups were not significantly different.

Regarding feedback duration (see Table 4), participants from all groups exhibited reduction in their feedback duration in the Exp-2 phase relative to the Pre-Eval phase. This reduction was most pronounced in **G1** ($\beta = -40.800$, $p < 0.001$; a reduction of approx. 40 seconds), followed by **G3** ($\beta = 17.558$, $p = 0.027$; a reduction of approx. 23 seconds), then **G2** ($\beta = 28.017$, $p < 0.001$; a reduction of approx. 13 seconds).

Table 4 The mixed-effect linear regression results for **feedback quality** and **feedback duration** between Pre-Eval and Exp-2 phases

	Feedback Quality				Feedback Duration			
Predictors	β	**SE**	**t value**	**p-value**	β	**SE**	**t value**	**p-value**
Intercept	1.802	0.131	13.707	< 0.001***	118.595	13.902	8.531	< 0.001***
Group								
G2	-0.237	0.157	-1.516	0.135	-42.202	18.928	-2.230	0.030*
G3	-0.139	0.159	-0.878	0.384	-32.476	19.189	-1.692	0.096
Phase								
Exp-2	-0.078	0.049	-1.581	0.114	-40.800	5.528	-7.381	< 0.001***
Interactions								
Exp-2 × G2	0.087	0.070	1.246	0.213	28.017	7.818	3.584	< 0.001***
Exp-2 × G3	0.105	0.071	1.485	0.138	17.558	7.925	2.216	0.027*

Note: SE – Standard Error; *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

The results for our RQ3 suggested that the natural-language assessment insights from GenAI were to some extent facilitating educators' establishment of more effective assessment practices that led to the most improvements in grading correctness and feedback quality among the three assessment conditions. However, such assessment practices could not promote more efficient grading and feedback provision compared to cases without additional support as educators likely needed to spend additional time in digesting and making use of the AI-powered insights. Meanwhile, the assessment practices established from prior exposure to important-word insights from BERT also enabled more improvements in grading correctness and feedback provision compared to the condition with no extra support, but at the cost of significantly greater time contributions in both grading and feedback provision.

5 Discussion

Our study contributed to bridging the gap in existing literature regarding a lack of knowledge on effective approaches to leverage GenAI in supporting educators' assessment practices (DiSabito et al., 2025). We designed a human study involving educators to engage in multiple phases of assessment activities to examine the effects of natural-language assessment insights from GenAI on educators' current and future assessment practices.

Our results yielded mix findings for the established hypotheses. For Hypothesis 1 (H1), relative To the condition where no insights had been made available, 1) the important-word insights deemed effective in previous research (Zeng et al., 2022) actually resulted in worse assessment performance in our context; 2) the natural-language insights led to more effective assessment (i.e., better grading correctness and feedback quality) but could not scale up efficiency. For Hypothesis 2 (H2), our findings suggested that the natural-language insights triggered more effective but less efficient assessment than the important-word insights. For Hypothesis 3 (H3), we noted that educators' prior exposure to AI-powered insights led to more effective but less efficient assessment practices. Due to the heterogeneity in our findings regarding the impacts of AI-powered insights on human assessment performance, our established hypotheses could not be fully accepted. Therefore, we further interpret our findings according to the RQs, as discussed below.

Regarding the impacts of natural-language insights during educators' grading process (RQ1), our findings contributed empirical evidence of the promises that such GenAI-enabled assessment insights hold to further scale up educators' grading accuracy (as shown in Table 1), resonating with claims made in pertinent piloting research (Xiao et al., 2024). Nevertheless, additional information introduced during grading naturally demands prolonged grading duration (reflected by the longer grading time of **G2** and **G3** relative to **G1**) as educators needed to contribute extra efforts to utilise such information (e.g., reading the GenAI insights). However, despite the fact that educators with important-word insights from BERT achieved higher reduction in grading duration compared to educators with natural-language insights from GenAI (approx. 10 seconds reduction vs. 4 seconds reduction, as shown in Table 1), the natural-language insights were subjectively rated to be more efficient in streamlining educators' decision-making during grading (52.6% positive ratings vs. 83.3% positive ratings, as shown in Fig. 7). These seemingly contradictory results may stem from the lower cognitive requirement to utilise the natural-language insights compared to the important-word insights (Block et al., 2010), leading to assessment insights in the form of natural language to be more preferable by educators to augment their grading practices. However, while educators commonly rated the natural-language insights highly, our observed higher improvement for grading correctness was not statistically significant, suggesting that the efficacy of such insights in supporting educators' grading of simple assessment tasks may be less pronounced.

With respect to feedback provision (RQ2), only educators who received GenAI-enabled natural-language assessment insights (**G3**) showed an improvement in the quality of their feedback alongside a reduction in feedback duration; whereas the remaining educators, despite shortened feedback duration, exhibited declines in feedback quality. The reduced feedback quality and duration for **G1** and **G2** may be subjected to educators' diminished capacity in feedback provision as a result of cognitive-intensive multitasking (Yuan & Zhong, 2024; Li et al., 2024). As grading and feedback provision both entail cognitive processes (Orrell, 2007), repetitive engagement in such activities can lead to depletion of mental resources (Ackerman, 2011), making it challenging for educators to maintain high performance in both tasks (Pashler, 1994; Cao & Liu, 2013; Gathmann et al., 2015), especially in cases where additional support is unavailable / ineffective. On the other hand, the natural-language insights enabled by GenAI (**G3**) deliver assessment support in a more accessible and informative manner that underscores the merits and shortcomings within students' answers, alleviating educators' cognitive workloads for justifying their grading decisions (Orrell, 2007), and thereby scaffolding educators' formulation of feedback for these answers efficiently (as also reflected by educators' more positive ratings of the GenAI insights in feedback provision). Yet, it is worth noting that the improvement in feedback quality facilitated by the natural-language insights was quantitatively modest.

While studies from broader educational research (e.g., Fan et al., 2025;Stadler et al., 2024) had highlighted a potential for reduced criticality in individuals' analytical skills as a result of leveraging GenAI to complete tasks, our findings suggested otherwise for educators' assessment practices. We noted that educators exposing to AI-powered insights (RQ3) exhibited certain learning effects which improved their grading correctness and feedback quality. This probably owes to only presenting the assessment insights without revealing the predicted scores, triggering educators' in-depth interpretation of the presented insights which contributes to their learning of effective assessment methods. However, the facilitation did not extend to their efficiency. One plausible explanation is that the AI-powered insights led educators to follow a particular assessment methodology. Educators previously exposed to BERT-enabled important-word insights may stick to the assessment approach of locating important text segments within students' answers before arriving at grading decisions and transforming such information into feedback for students. However, while these important words may be effective for educators' assessment, identifying and leveraging them during assessment may not be straightforward and can be time-consuming. Meanwhile, educators with prior exposure to GenAI insights may learn from the natural-language explanations to establish a set of grading rules by which they assess students' answers. While these rules are more effective, matching these rules to students' answers and converting such outcomes into feedback can take additional time. Our results offer empirical evidence to suggest that engagement with AI-powered insights, especially the natural-language ones enabled by GenAI, during assessment can promote the establishment of more effective assessment practices in future assessment tasks.

Our study offers critical insights that contribute to the establishment of mechanisms for human-AI collaborative assessment, advancing the discourse on AI-assisted methodologies as explored in the existing literature. For educational researchers, these findings highlight the potential of GenAI to serve as an effective auxiliary source of information to support educators' assessment. However, as the GenAI-enabled natural-language insights explored in our study still require extra time for educators to read and utilise, educational researchers should further investigate into optimising such natural-language insights so as to improve assessment quality while maintaining efficiency. Furthermore, since the effectiveness of the natural-language insights in our study was somewhat constrained by the simplicity of the assessment tasks, educational researchers should extend this line of research to more challenging tasks (e.g., essays, coding) to evaluate the effectiveness of such insights in augmenting educators' assessment practices in other educational scenarios. In addition, the modest improvement in feedback quality enabled by the natural-language insights calls for the exploration of other potentially more effective human-AI collaborative feedback approach, such as instructing GenAI to draft the feedback based on the natural-language assessment rationales and involving educators to revise the feedback, or vice versa. For educators, our findings illustrated the promises for GenAI-enabled natural-language insights to complement educators' innate grading capabilities, potentially boosting grading consistency. Educators may leverage such insights to support their assessment and potentially alleviate the cognitive requirements during assessment. However, educators should be mindful not to overly rely on GenAI insights, and should retain their own judgement on these insights. For educational technology developers, this study drives the potential development of assessment tools that integrate GenAI assessment insights to support educators' assessment practices.

6 Conclusion, Limitations and Future Work

In conclusion, our study provides new insights into the adoption of GenAI technologies within educational assessment practices, specifically examining how GenAI-enabled natural-language assessment insights influence grading accuracy, feedback provision, and educators' assessment behaviours. Compared to both the control group with no AI support (**G1**) and the group receiving AI-powered important-word insights (**G2**), the group leveraging GenAI-enabled natural-language insights (**G3**) demonstrated the largest improvement in grading correctness (log-odds increase of 0.556 relative to **G1**). While this improvement did not reach statistical significance, the notable increase suggested a promising potential for natural-language insights to enhance grading accuracy beyond what could be achieved with important-word or no AI support. Notably, while exposure to AI-powered insights led to improvements in feedback quality for both **G2** ($\beta = 0.065$, $p = 0.366$) and **G3** ($\beta = 0.190$, $p = 0.010$), relative to **G1**, the GenAI-enabled natural-language insights enabled significantly higher quality feedback. The superior effectiveness of the natural-language insights resonated with the fact that these insights were rated significantly higher by educators compared to the important-word insights, leading to educators' greater willingness to adopt the natural-language insights in their actual teaching practices. More importantly, our study provides initial empirical evidence that prior exposure to AI-powered insights can foster the development of more effective assessment practices, even in subsequent tasks without AI support. This is demonstrated by the higher grading accuracy and feedback quality achieved by both **G2** and **G3** compared to **G1**, with the most pronounced improvements observed among **G3** participants who had engaged with natural-language insights.

Whereas our research involves experienced human graders from crowdsourcing platforms instead of from authentic pedagogical contexts, we anticipate the identified positive effects to be more pronounced under real-world assessment settings as educators are likely to have deeper subject matter expertise, better understanding of learning outcomes, and more awareness of contextual circumstances (e.g., student progress). However, while findings from our study suggest mostly positive effects of GenAI-enabled natural-language insights in fostering more effective assessment from educators, it is worth noting that our experiments only leverage assessment insights from correct GenAI assessment. Despite the potential effects of these insights in alleviating the cognitive requirement associated with assessment, it still poses concerns about "metacognitive laziness" (Fan et al., 2025), where individuals may blindly trust the GenAI-enabled insights without in-depth evaluation to their appropriateness and correctness. When presented faulty insights from incorrect GenAI assessment, such laziness can potentially lead to educators' establishment of inaccurate assessment approaches that exacerbate concerns in assessment reliability. Hence, as GenAI assessment is not impeccable, future work should explore how assessment insights from incorrect GenAI assessment influence human assessment to contribute a more holistic picture of potential benefits and challenges of AI-assisted assessment. Furthermore, as some of our reported results only suggest a potential statistical trend (i.e., a p-value approaching the significance threshold), to offer more concrete evidence, future research may replicate our study in a larger scale and include tasks from more complex contexts, where the availability of additional support plays a more crucial role.

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Data Availability The student response datasets used in our study is publicly available and can be accessed from https://www.kaggle.com/competitions/asap-sas. The data generated during and/or analysed during the current study (i.e., the assessment data from the study participants) are available from the corresponding authors upon reasonable request.

Declarations

Conflicts of interest There is no known conflict of interest for this paper.

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